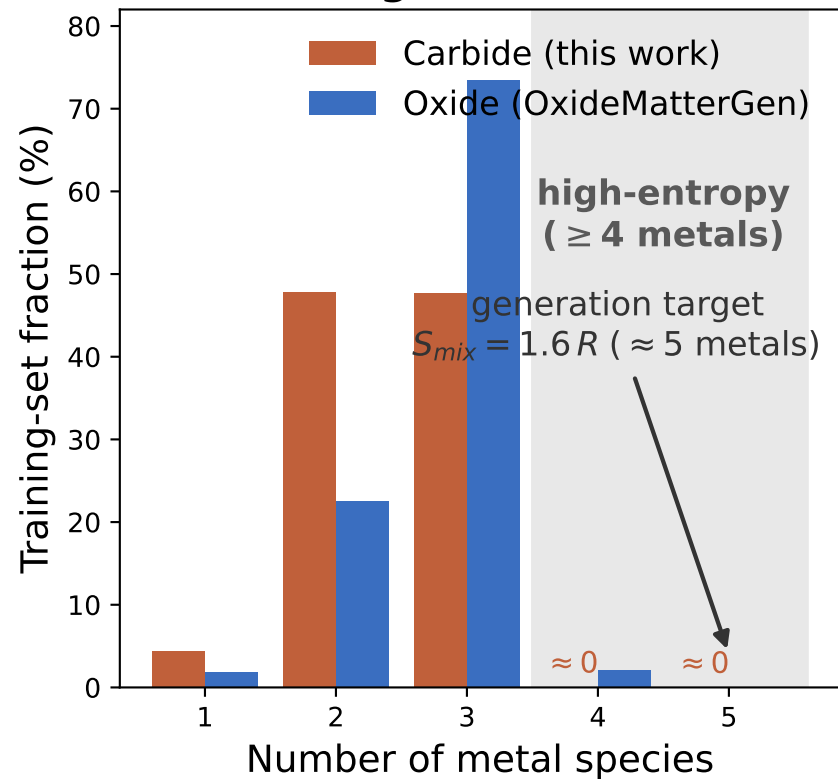


Why the generative model cannot make high-entropy carbides

(a) Training-data distribution

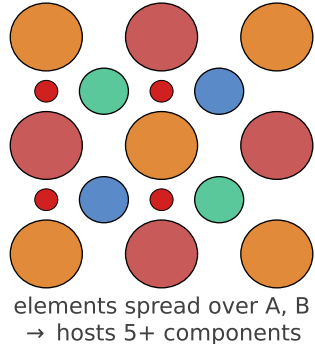
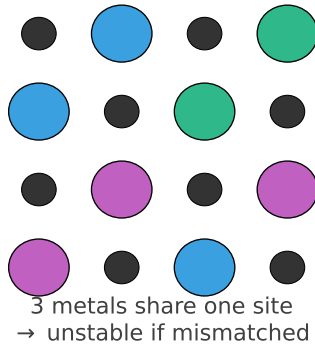


(b) Why? Sublattice structure

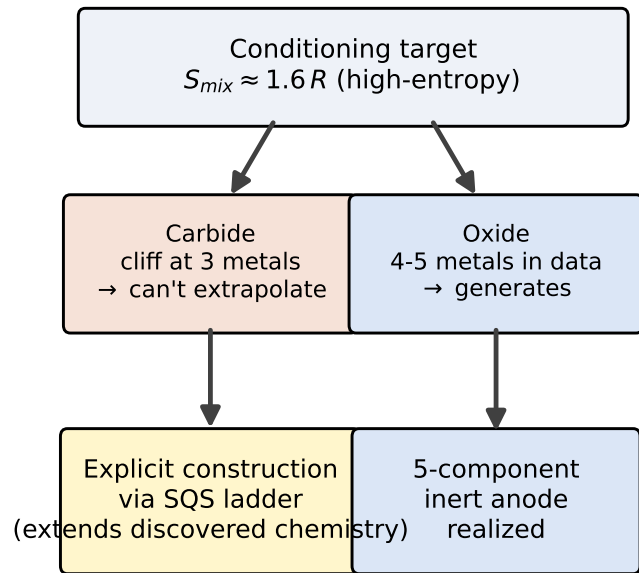
Rocksalt carbide **Perovskite oxide**

1 metal sublattice

2 sublattices (A, B)



(c) Outcome



Carbide ≥ 4 metals: 0.04% vs Oxide ≥ 4 metals: 2.1% (incl. 5-6 metals)